

JINWANG WANG

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EDUCATION

Wuhan University, Wuhan, China

Sep. 2016 – Present

Ph.D. Candidate in Communication and Information System

Lanzhou University, Lanzhou, China

Sep. 2012 – Jul. 2016

B.S. in Communication Engineering, GPA 4.8/5.0, RANK 2nd of 72

RESEARCH EXPERIENCE

Signal Processing Lab, Wuhan University

Jul. 2016 – Present

Research Internship for object detection, localization and recognition, Advised by Prof. Wen Yang

- Arbitrary-Oriented Object Detection in Aerial Images (Oct. 2018 - Now)
 - Designed a novel arbitrary-oriented object detection framework for aerial images in DOTA dataset adapted from Mask R-CNN
- The Intelligent Unmanned Aerial Manipulator for Dangerous Environment (Nov. 2017 - Oct. 2018)
 - Built a bottle dataset (UAV-BD) collected from UAV images with the rotation bounding box
 - Proposed Rotation-SqueezeDet to detect and localize oriented objects, reached about 16FPS running speed on NVIDIA Jetson TX2, achieved 76.5% detection accuracy and 1cm location accuracy
 - Designed an autonomous Rotation-Aware unmanned aerial grasping system based on the Rotation-SqueezeDet, achieved 80% success rate
- The Intelligent Unmanned Aircraft System for Wilderness Search and Rescue (Nov. 2016 - Sep. 2017)
 - Collected about 20,000 UAV images to build a person dataset (UAV-PP)
 - Adapted detection algorithms YOLOv2, SSD and Faster R-CNN for person detection and chose the SSD as the detection algorithm due to it's balance of accuracy and speed, the detection accuracy achieved 88.92%
- UAV Detection and Automatic Tracking System (Jun. 2016 - Nov. 2016)
 - Designed UAV detection and automatic tracking system, including designed the PTZ control algorithm, implemented real-time tracking algorithm KCF to track UAV with low-speed arbitrary flight
 - Collected about 3,000 UAV images of DJI Phantom 3 for training the detection model of HOG+SVM, achieved 78.6% detection accuracy

The Bachelor's Degree Thesis, Lanzhou University

Oct. 2015 – Jul. 2016

Title: Study of Superpixel Segmentation for Unmanned Aerial Vehicle Aerial Image, Advised by Prof. Wen Yang

- Adapted the superpixel segmentation algorithms including SLIC, SEEDS, WaterSheds, and MeanShift for UAV aerial image segmentation
- Implemented an instance segmentation algorithm based on SLIC, spectral clustering and Region Adjacency Graph (RAG)

Undergraduate Researcher with the Support of Chun-Tsung Endowment Fund, Lanzhou University

Research Internship for programmable DC electric metamaterials, Advised by Prof. Zhonglei Mei

Mar. 2015 – Mar. 2016

- Developed a programmable DC electric metamaterial using MCP42010 and MSP430, achieved the DC electric invisibility cloak, the DC electric concentrator and the DC electric illusion in one circuit board
- Designed a multilayer absorbing metamaterial by genetic algorithm, improved about 10% absorbing rate compared with normal absorbing material

HONORS AND AWARDS

National Scholarship	2018
1 st Prize & Best Paper Award, National Graduate Electronic Design Contest	2018
Finalists & 6 th Place, ICRA 2018 DJI RoboMaster AI Challenge	2018
National Scholarship	2017
2 nd Prize, National Graduate Electronic Design Contest	2017
2 nd Place of Breakthrough Competition, National “Gaofen Cup” IntelliSense Technology Contest for UAV	2017
3 rd Place of Comprehensive Competition, National “Gaofen Cup” IntelliSense Technology Contest for UAV	2017
Finalists & 6 th Place, ICRA2017 DJI RoboMaster Mobile Manipulation Challenge	2017
1 st Prize, The Scholarship for Excellent Freshman, <i>Wuhan University</i>	2016
Outstanding Graduates Awards, <i>Lanzhou University</i>	2016
Excellent Graduation Thesis, <i>Lanzhou University</i>	2016
Honorable Mention, Mathematical Contest in Modeling Certificate of Achievement	2016
Chun-Tsung Scholar, <i>Chun-Tsung Endowment (1%)</i>	2016
National Scholarship for Encouragement (1%)	2015
3 rd Prize, National “Challenge Cup” Undergraduate Curricular Academic Science and Technology	2015
Outstanding Award, Undergraduate Electronic Design Contest of Gansu Province	2015
Merit Student in Gansu Province	2015
2 nd Prize, National Undergraduate Mathematical Contest in Modeling	2014
National Scholarship (1%)	2014
National Scholarship for Encouragement (1%)	2013

PUBLICATIONS

Journal Papers

- Shijie Lin*, **Jinwang Wang***, Wen Yang, Guisong Xia. Toward Autonomous Rotation-Aware Unmanned Aerial Grasping. *IEEE Robotics and Automation Letters (RA-L)*, 2018. (Submitted)
- Huai Yu, **Jinwang Wang**, Yu Bai, Wen Yang, Gui-Song Xia. Analysis of Large Scale UAV Images using A Multi-scale Hierarchical Representation. *Geo-spatial Information Science*, 2018, 21(1):33-44.

Conference Papers

- **Jinwang Wang**, Wei Guo, Ting Pan, Huai Yu, Lin Duan, Wen Yang. Bottle Detection in the Wild Using Low-Altitude Unmanned Aerial Vehicles. *2018 21st International Conference on Information Fusion (FUSION)*, Cambridge, 2018, pp. 439-444.
- Fang Xu, Huai Yu, **Jinwang Wang**, Wen Yang. Accurate Registration of Multitemporal UAV Images Based on Detection of Major Changes. *2018 21st International Conference on Information Fusion (FUSION)*, Cambridge, 2018, pp. 1480-1485.
- Xu Lei, **Jinwang Wang**, Kaimin Fu, Huai Yu, Wen Yang. Accurate Object Matching for UAV Imagery Using Multi-scale Best-buddies Similarity. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Fort Worth, TX, 2017, pp. 3341-3344.
- Huai Yu, Shijie Lin, **Jinwang Wang**, Kaimin Fu, Wen Yang. An Intelligent Unmanned Aircraft System for Wilderness Search and Rescue. *International Micro Air Vehicles Conference and Flight Competition (IMAV)*, Toulouse, 2017, pp. 129-135.

TEACHING

- Teaching Assistant (TA) with *Prof. Xin Xu*, **Advanced Digital Signal Processing (ADSP)**, Wuhan University, Fall 2018

SKILLS

- **Programming:** Python, C/C++, MATLAB
- **Tools:** PyTorch, OpenCV, ROS, TensorFlow, Caffe, Ubuntu, L^AT_EX
- **Language:** Chinese (native), English (CET-6)